Heart Failure Prediction

The purpose of this research paper is to provide a comprehensive, step-by-step guide for conducting a complete statistical workflow using the Isalos Analytics Platform.

Firstly, we conduct **descriptive statistics** to summarize the main characteristics of the patient cohort (demographics and clinical measurements). Secondly, we perform a **Multivariate Analysis of Covariance (MANCOVA)** to simultaneously assess how categorical risk factors influence multiple correlated cardiovascular outcomes, while controlling for the confounding effect of age.

Isalos version used: 2.1.5

The dataset used in this analysis is the **Heart Disease Dataset** (UCI), available on Kaggle.

Source: Heart Disease UCI

This dataset simulates real-world healthcare records and includes patient-level information for over 900 individuals. It contains demographic details (age, sex), clinical measurements (resting blood pressure, cholesterol, maximum heart rate), and lifestyle/symptom indicators (chest pain type, exercise angina). It is widely used for predictive modelling and risk assessment for cardiovascular disease.

Part 1 : Descriptive Statistics

The purpose of this section is to summarise the main characteristics of the patient cohort, including central tendency, dispersion, and distribution shape of the key clinical variables.

Step 1: Import Data from file

Firstly, launch the Isalos Analytics Platform (version 2.1.2). Then, in the left spreadsheet panel, create a new tab by clicking **File** and choose **Open** to load the heart dataset (or simply copy-paste the data from your spreadsheet).

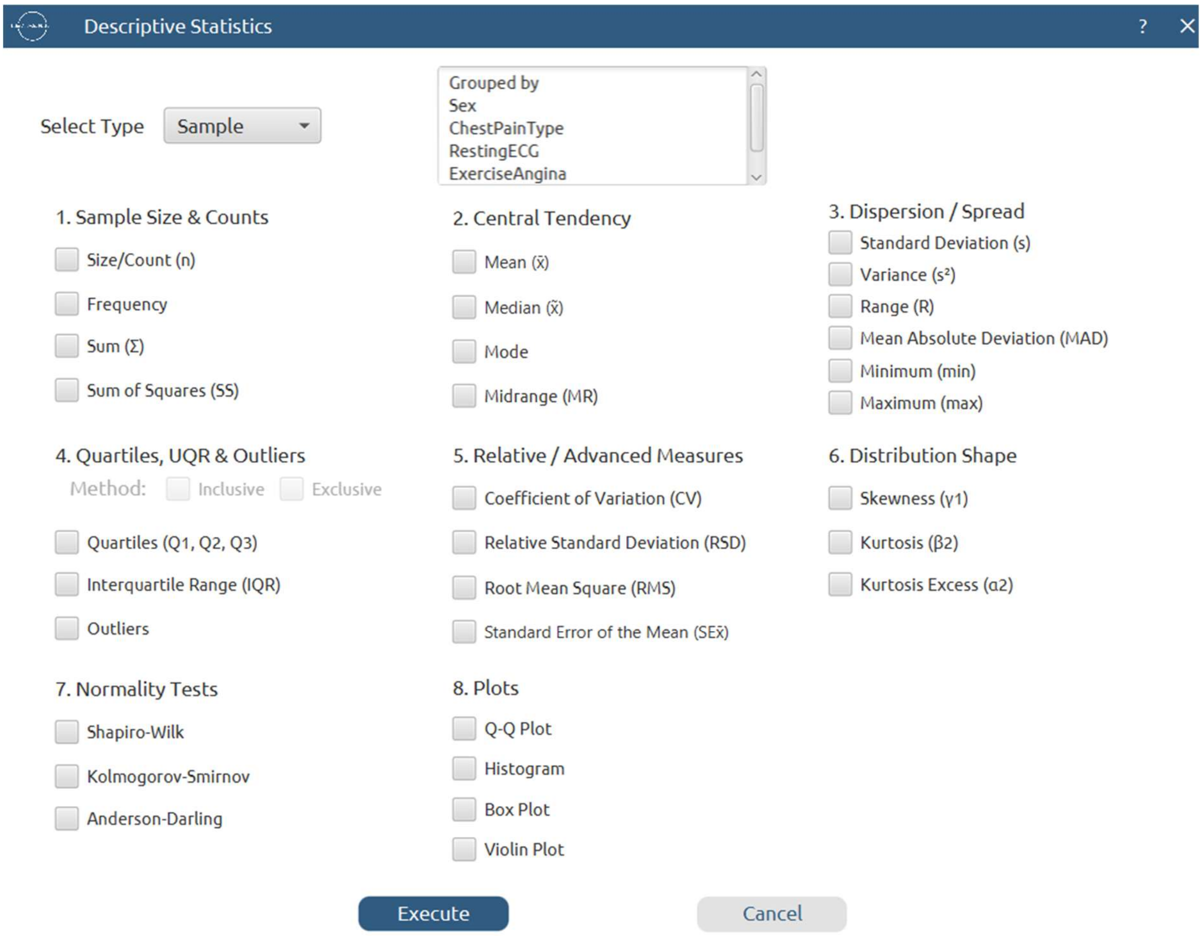
Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDiseas
40	M	ATA	140	289	0	Normal	172	N	0	Up	0
49	F	NAP	160	180	0	Normal	156	N	1	Flat	1
37	M	ATA	130	283	0	ST	98	N	0	Up	0
48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
54	M	NAP	150	195	0	Normal	122	N	0	Up	0
39	M	NAP	120	339	0	Normal	170	N	0	Up	0
45	F	ATA	130	237	0	Normal	170	N	0	Up	0
54	M	ATA	110	208	0	Normal	142	N	0	Up	0
37	M	ASY	140	207	0	Normal	130	Y	1.5	Flat	1
48	F	ATA	120	284	0	Normal	120	N	0	Up	0
37	F	NAP	130	211	0	Normal	142	N	0	Up	0
58	M	ATA	136	164	0	ST	99	Y	2	Flat	1
39	M	ATA	120	204	0	Normal	145	N	0	Up	0
49	M	ASY	140	234	0	Normal	140	Y	1	Flat	1
42	F	NAP	115	211	0	ST	137	N	0	Up	0

Step 2: Access the Descriptive Statistics function

Navigate to the descriptive statistics module through the main menu by choosing:

Statistics → Basic Statistics → Descriptive Statistics

Then, you will see the descriptive statistics configuration window:



Step 3: Configure parameters

In the descriptive statistics window, set the following parameters:

1. Sample Size & Counts
    a. Size / Count (n)
    b. Frequency
    c. Sum
    d. Sum of Squares

2. Central Tendency
    a. Mean
    b. Median
    c. Mode
    d. Midrange

3. Dispersion / Spread
    a. Standard Deviation

   b. Variance
   c. Range
   d. Mean Absolute Deviation
   e. Minimum
   f. Maximum

4. Quartiles, UQE & Outliers
   a. Quartiles (Q1,Q2,Q3)
   b. Interquartile Range (IQR)
   c. Outliers

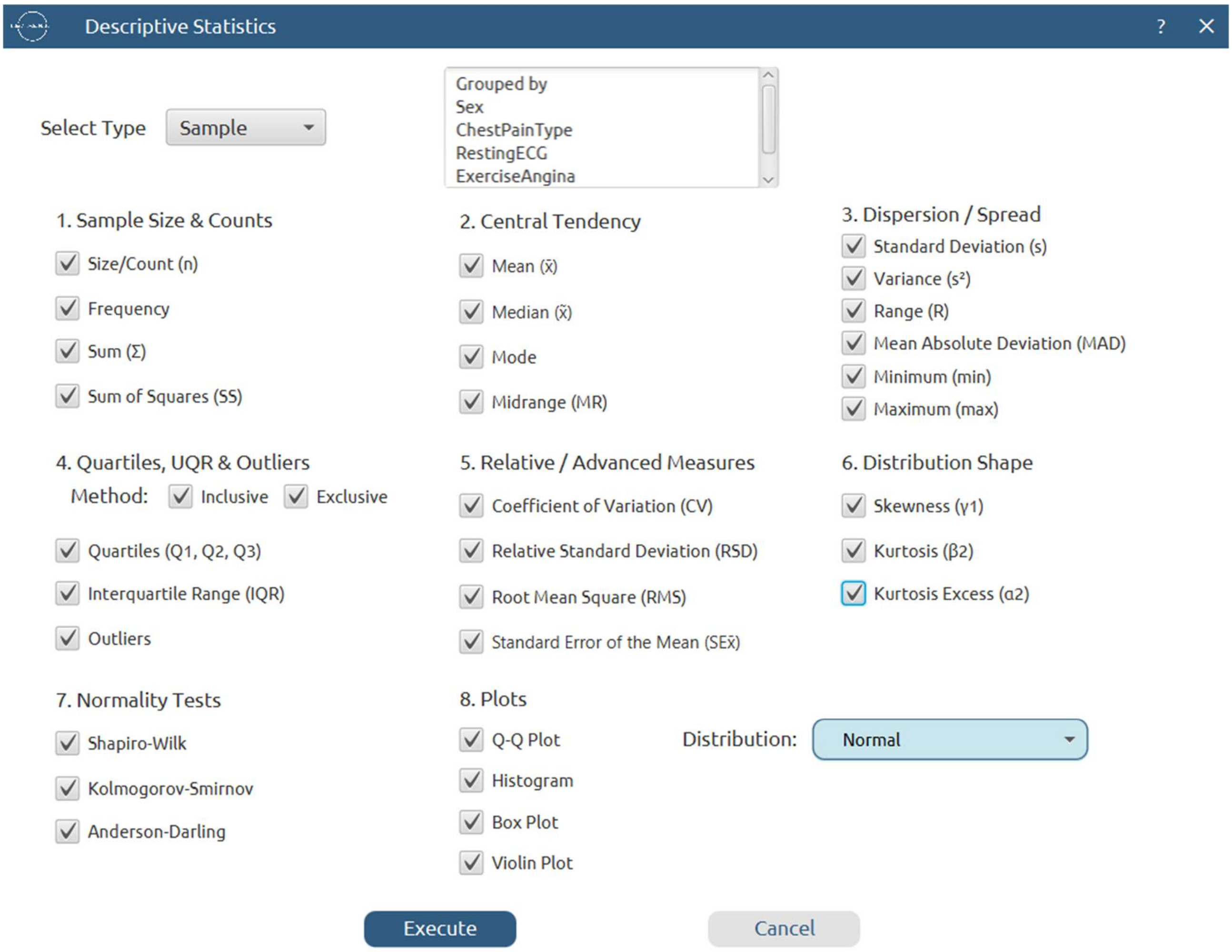
5. Relative / Advanced Measures
   a. Coefficient of Variation
   b. Relative Standard Deviation
   c. Root Mean Square
   d. Standard Error of the Mean

6. Distribution Shape
   a. Skewness
   b. Kurtosis
   c. Kurtosis excess

7. Normality Tests
   a. Shapiro-Wilk
   b. Kolmogorov-Smirnov
   c. Anderson-Darling

8. Plots
   a. Q-Q Plot
   b. Histogram
   c. Box Plot
   d. Violin Plot

The preview below displays the multivariate model:



After configuring all parameters:

- Review all selections to ensure accuracy.
- Click the **Execute** button to run the descriptive analysis.
- Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components: (some of them)

Sex	ChestPainType	RestingECG	ExerciseAngina	ST_Slope	Age : Size/Count	Age : Sum	Age : Sum of Squares	Age : Mean	Age : Median	Age : Mode	Age : Mid Range
M	ATA	Normal	N	Up	64.0	3028.0	4485.75	47.3125	48.5	52.0, 54.0	45.5

Age : Standard Deviation	Age : Variance	Age : Range	Age : Mean Absolute Deviation	Age : Minimum	Age : Maximum	Age : Quartiles - Inclusive	Age : Quartiles - Exclusive	Age : Interquartile Range - Inclusive	Age : Interquartile Range - Exclusive
8.4381503	71.2023810	33.0	7.1894531	29.0	62.0	Q1: 41.000000, Q2: 48.500000,Q3: 54.000000	Q1: 41.000000, Q2: 48.500000,Q3: 54.000000	13.0	13.0

Age : Coefficient Of Variation	Age : Relative Standard Deviation	Age : Root Mean Square	Age : Standard Error Of Mean	Age : Skewness	Age : Kurtosis	Age : Kurtosis Excess	Age : Shapiro-Wilk: W	Age : Shapiro-Wilk: p-value	Age : Shapiro-Wilk: Prob	Age : Shapiro-Wilk: Decision	Age : Anderson-Darling: A²
0.1783493	17.8349280	48.047503	1.0547688	-0.3193837	2.2532517	-0.7467483	0.9660686	0.0754195	0.9245805	Fail to Reject Null Hypothesis	0.1169780

Age : Anderson-Darling: p-value	Age : Anderson-Darling: Prob	Age : Anderson-Darling: Decision	Age : Kolmogorov-Smirnov: D	Age : Kolmogorov-Smirnov: p-value	Age : Kolmogorov-Smirnov: Prob	Age : Kolmogorov-Smirnov: Decision
0.3199129	0.6800871	Fail to Reject Null Hypothesis	0.7290598	0.0545066	0.9454934	Fail to Reject Null Hypothesis

RestingBP : Quartiles - Inclusive	RestingBP : Quartiles - Exclusive	RestingBP : Interquartile Range - Inclusive	RestingBP : Interquartile Range - Exclusive	RestingBP : Outliers - Inclusive	RestingBP : Outliers - Exclusive	RestingBP : Coefficient Of Variation	RestingBP : Relative Standard Deviation	RestingBP : Root Mean Square	RestingBP : Standard Error Of Mean	RestingBP : Skewness
Q1: 120.000000, Q2: 129.000000,Q3: 140.000000	Q1: 120.000000, Q2: 129.000000,Q3: 140.000000	20.0	20.0	190.0	190.0	0.1233311	12.3331060	129.820489	1.9865454	0.8961159

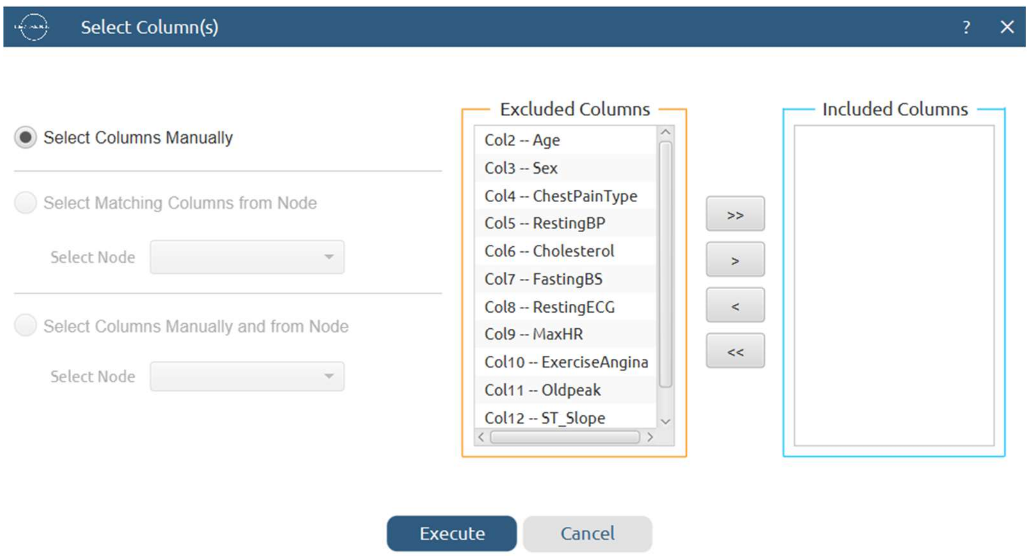
RestingBP : Kurtosis	RestingBP : Kurtosis Excess	RestingBP : Shapiro-Wilk: W	RestingBP : Shapiro-Wilk: p-value	RestingBP : Shapiro-Wilk: Prob	RestingBP : Shapiro-Wilk: Decision	RestingBP : Anderson-Darling: A²	RestingBP : Anderson-Darling: p-value	RestingBP : Anderson-Darling: Prob	RestingBP : Anderson-Darling: Decision
5.6138602	2.6138602	0.9245611	0.0007749	0.9992251	Reject Null Hypothesis	0.1636062	0.0577612	0.9422388	Fail to Reject Null Hypothesis

RestingBP : Size/Count	RestingBP : Sum	RestingBP : Sum of Squares	RestingBP : Mean	RestingBP : Median	RestingBP : Mode	RestingBP : Mid Range	RestingBP : Standard Deviation	RestingBP : Variance	RestingBP : Range	RestingBP : Mean Absolute Deviation	RestingBP : Minimum	RestingBP : Maximum
64.0	8247.0	15911.734375	128.859375	129.0	120.0	144.0	15.8923633	252.5672123	92.0	11.796875	98.0	190.0

To get the plots we must follow the next path:

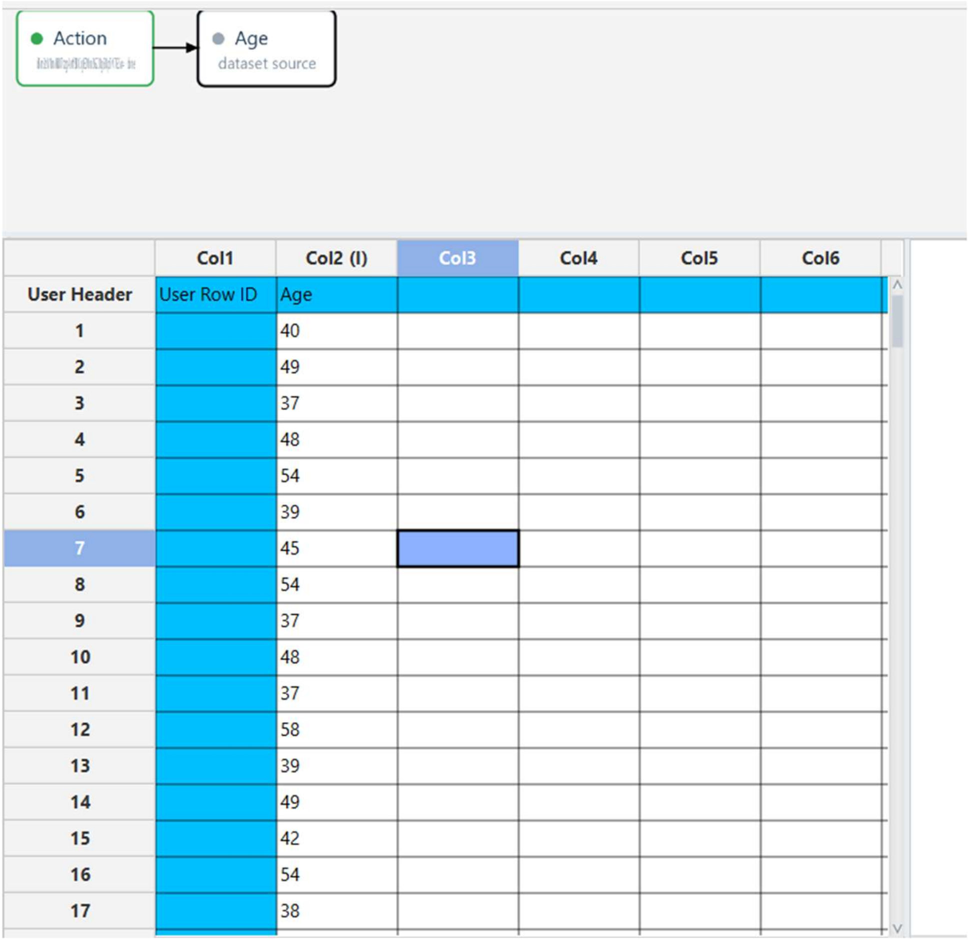
Data transformation → Data manipulation → Select Column

Then you get the below window:



Select the desired variables from the **Excluded Columns** list and move them to the **Included Columns** list. Once the required columns have been selected, click **Execute**.

Next, create a new tab by clicking the **"+"** button and give it an appropriate name. Then, right-click on the new tab and select **Import from Spreadsheet**. Click **Execute** again. The following window will appear:



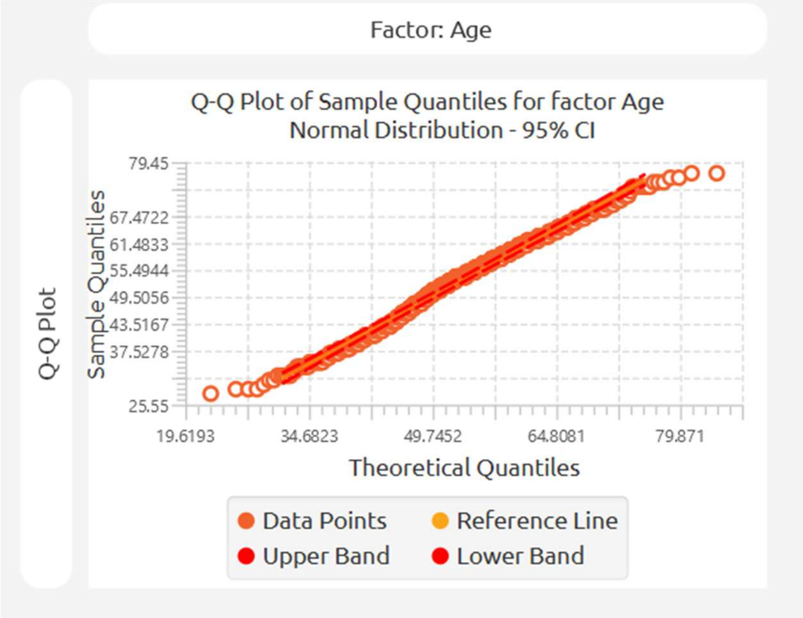
Next, follow the same path as before:

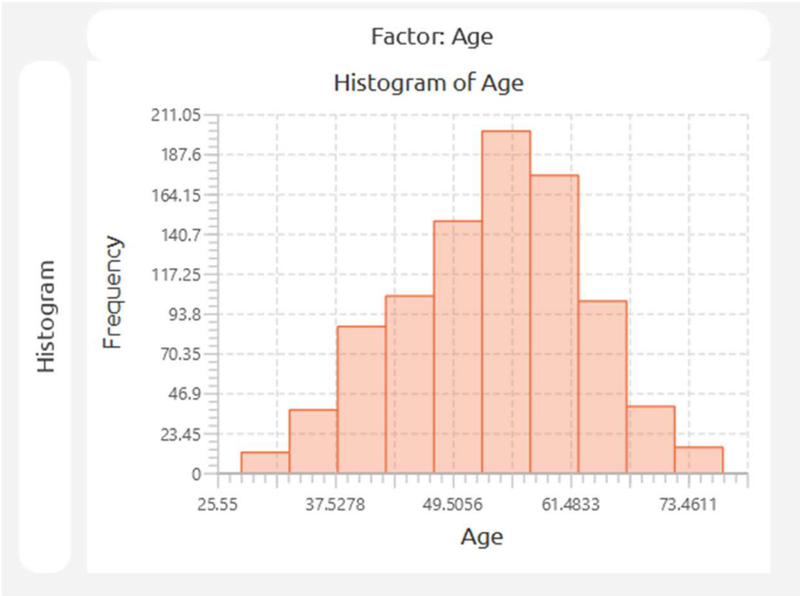
Statistics → Basic Statistics → Descriptive Statistics

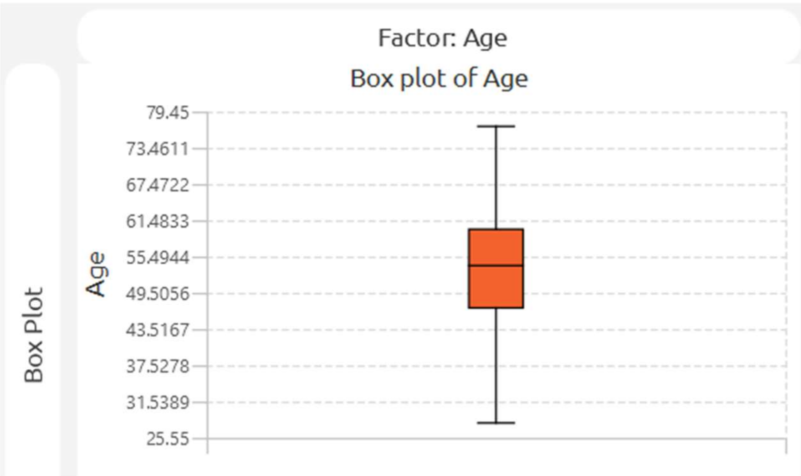
Select the type of plot you want to generate and click **Execute**.

The examples below show three commonly used descriptive plots:

- **Q–Q Plot**
- **Histogram**
- **Boxplot** (with **Age** used as the grouping factor)


Factor: Age
Q-Q Plot
Q-Q Plot of Sample Quantiles for factor Age
Normal Distribution - 95% CI
Sample Quantiles
79.45
67.4722
61.4833
55.4944
49.5056
43.5167
37.5278
25.55
19.6193
34.6823
49.7452
64.8081
79.871
Theoretical Quantiles
Data Points
Reference Line
Upper Band
Lower Band


Factor: Age
Histogram
Histogram of Age
Frequency
211.05
187.6
164.15
140.7
117.25
93.8
70.35
46.9
23.45
0
25.55
37.5278
49.5056
61.4833
73.4611
Age


Factor: Age
Box Plot
Box plot of Age
Age
79.45
73.4611
67.4722
61.4833
55.4944
49.5056
43.5167
37.5278
31.5389
25.55

Part 2 : Multivariate Analysis of Covariance (MANCOVA)

Having summarised the data, we now proceed to the multivariate modelling. MANCOVA extends ANCOVA by simultaneously modelling **multiple continuous dependent variables**, allowing us to assess how categorical factors and a continuous covariate jointly influence a set of interrelated cardiovascular outcomes.

For the purposes of this MANCOVA analysis, variables are classified as follows:

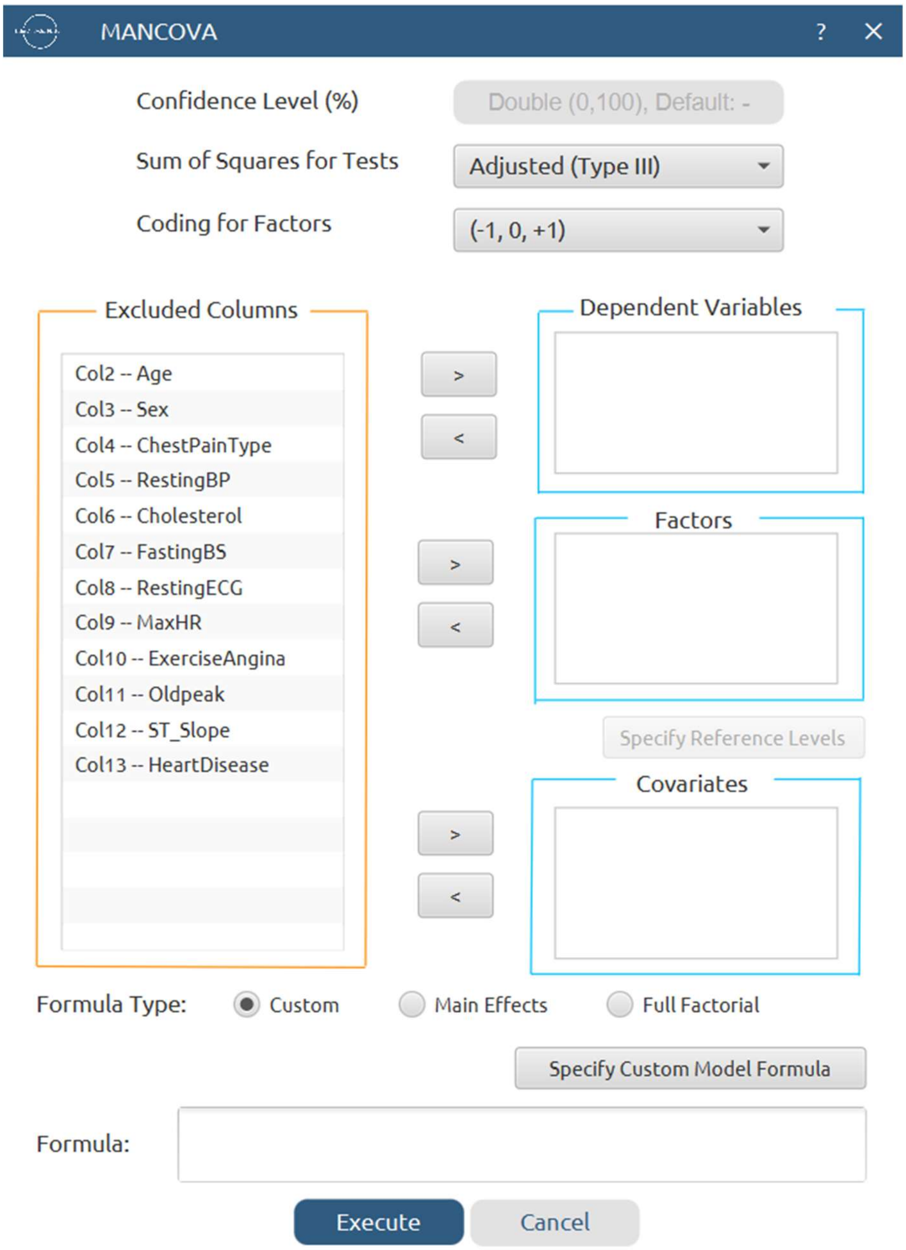
- **Dependent Variables (Responses):**
  - *Resting BP* (Resting Blood Pressure)
  - *Cholesterol* (Serum Cholesterol)
  - *Max HR* (Maximum Heart Rate)
    (These are continuous outcome measures to be predicted simultaneously).
- **Factors (Categorical Predictors):**
  - *Sex* (e.g., Male, Female)
  - *Chest Pain Type* (e.g., ATA, NAP, ASY, TA)
    (These categorical variables represent demographic/clinical groups whose multivariate effect is of primary interest).
- **Covariate (Continuous Predictor):**
  - *Age* (in years)
    (This continuous variable is controlled for to remove its confounding influence on the dependent variables).

Step 1: Access the MANCOVA function

Navigate to the MANCOVA analysis module through the main menu by choosing the following path:

Statistics → Analysis of (Co)Variance → MANCOVA

Then, you will see the MANCOVA configuration window:



Step 2: Configure MANCOVA parameters

The MANCOVA model can be specified as follows. The model is expressed as:

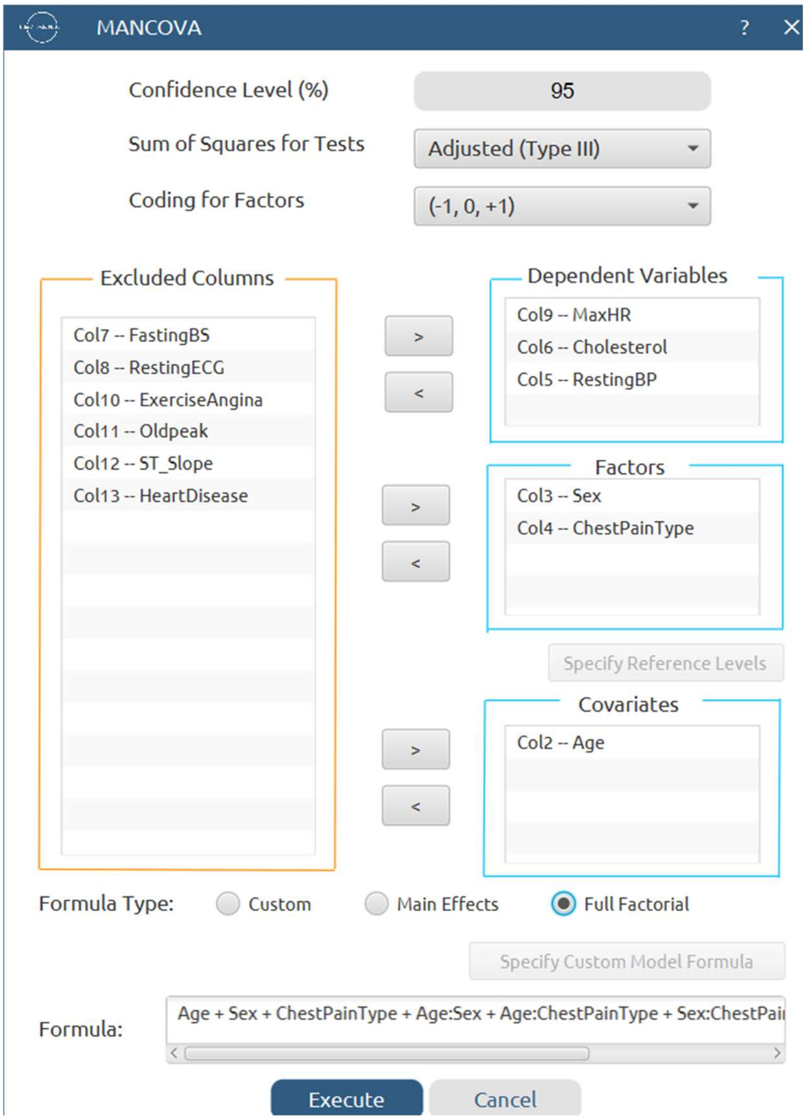
$$\mathbf{[RestingBP, Cholesterol, MaxHR] = \beta_0 + \beta_1(Age) + \beta_2(Sex) + \beta_3(ChestPainType) + \varepsilon}$$

This is a multivariate linear model where the three dependent variables are regressed onto the covariate (Age) and the two categorical factors. For this basic demonstration, we focus on the **main-effects model**.

In the MANCOVA configuration window, set the following parameters:

- Confidence Level (%) : 95
- **Dependent Variables:** Move the following to the ***Dependent Variables*** field:
  - *Resting BP*
  - *Cholesterol*
  - *Max HR*
- **Factors (Categorical Predictors):** Move the following to the ***Factors*** field:
  - *Sex*
  - *Chest Pain Type*
- **Covariates (Continuous Predictors):** Move the following to the ***Covariates*** field:
  - *Age*

Then, for **Formula Type**, choose **Full Factorial**. The formula preview will display the multivariate model as shown below:



After configuring all parameters:

- Review all selections to ensure accuracy.
- Click the **Execute** button to run the MANCOVA analysis.
- Wait for the computation to complete.

Step 3: MANCOVA Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

Source	Dependent Variable	DF	Adj SS	Adj MS	F-Value	P-Value	Significant
Age	MaxHR	1	29116.9552506	29116.9552506	60.1152455	0E-7	Yes
	Cholesterol	1	683.2826198	683.2826198	0.0607383	0.8053889	No
	RestingBP	1	10538.9541647	10538.9541647	32.7578599	0E-7	Yes
Sex	MaxHR	1	376.7100253	376.7100253	0.7777604	0.3780619	No
	Cholesterol	1	15146.7460250	15146.7460250	1.3464228	0.2462115	No
	RestingBP	1	3.4136502	3.4136502	0.0106105	0.9179800	No
ChestPainType	MaxHR	3	6373.0590626	2124.3530209	4.3859669	0.0044875	Yes
	Cholesterol	3	1237.1333192	412.3777731	0.0366570	0.9906078	No
	RestingBP	3	161.9233913	53.9744638	0.1677669	0.9181352	No
Age*Sex	MaxHR	1	795.1368804	795.1368804	1.6416500	0.2004285	No
	Cholesterol	1	40042.1216533	40042.1216533	3.5594196	0.0595294	No
	RestingBP	1	22.9428950	22.9428950	0.0713126	0.7894962	No
Age*ChestPainType	MaxHR	3	3381.1519056	1127.0506352	2.3269234	0.0732568	No
	Cholesterol	3	2205.3803347	735.1267782	0.0653468	0.9782120	No
	RestingBP	3	235.9153880	78.6384627	0.2444292	0.8653222	No
Sex*ChestPainType	MaxHR	3	365.4283909	121.8094636	0.2514894	0.8602993	No
	Cholesterol	3	13886.7717012	4628.9239004	0.4114738	0.7447994	No
	RestingBP	3	697.0315305	232.3438435	0.7221862	0.5388434	No

Age*ChestPainType	MaxHR	3	3381.1519056	1127.0506352	2.3269234	0.0732568	No
	Cholesterol	3	2205.3803347	735.1267782	0.0653468	0.9782120	No
	RestingBP	3	235.9153880	78.6384627	0.2444292	0.8653222	No
Sex*ChestPainType	MaxHR	3	365.4283909	121.8094636	0.2514894	0.8602993	No
	Cholesterol	3	13886.7717012	4628.9239004	0.4114738	0.7447994	No
	RestingBP	3	697.0315305	232.3438435	0.7221862	0.5388434	No
Age*Sex*ChestPainType	MaxHR	3	736.6682748	245.5560916	0.5069783	0.6775480	No
	Cholesterol	3	22293.1243371	7431.0414457	0.6605593	0.5764429	No
	RestingBP	3	483.0818047	161.0272682	0.5005154	0.6820098	No
Corrected Model	MaxHR	15	157539.8973976	10502.6598265	21.6839284	0.0	Yes
	Cholesterol	15	824646.8623591	54976.4574906	4.8869608	0E-7	Yes
	RestingBP	15	24129.5997278	1608.6399819	5.0000790	0E-7	Yes
Error	MaxHR	902	436885.7420359	484.3522639			
	Cholesterol	902	10147158.2574666	11249.6211280			
	RestingBP	902	290194.0691175	321.7229148			
Corrected Total	MaxHR	917	594425.6394336				
	Cholesterol	917	10971805.1198257				
	RestingBP	917	314323.66884				

Step 4 : Residual Analysis & Assumption's checks

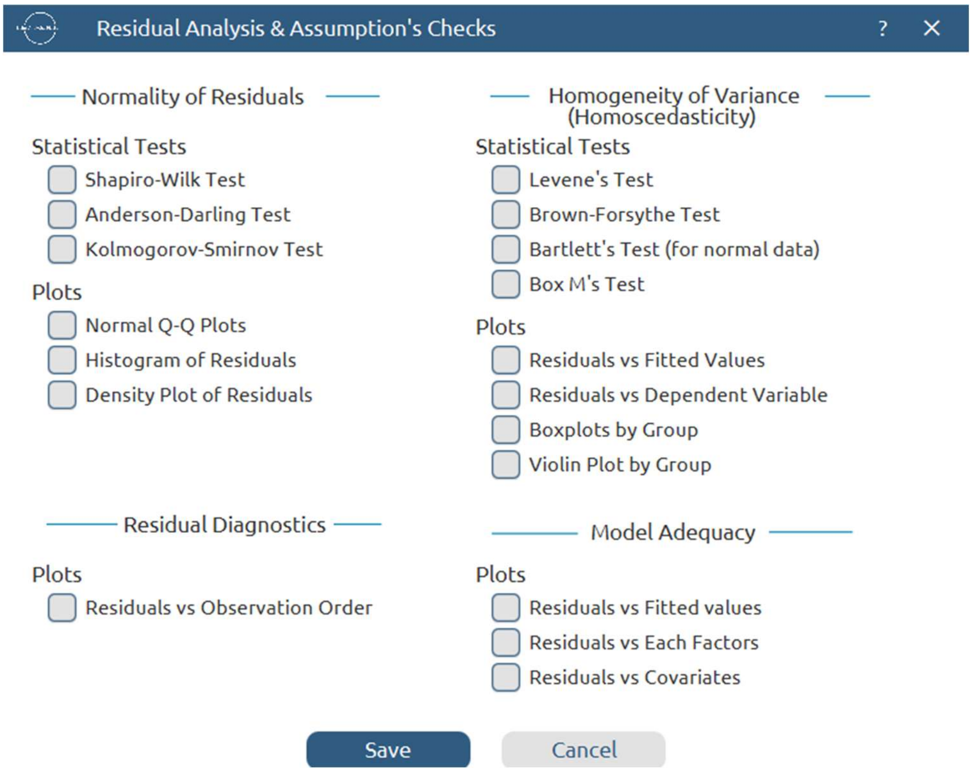
Choose

Statistics → Analysis of (Co)Variance → MANCOVA

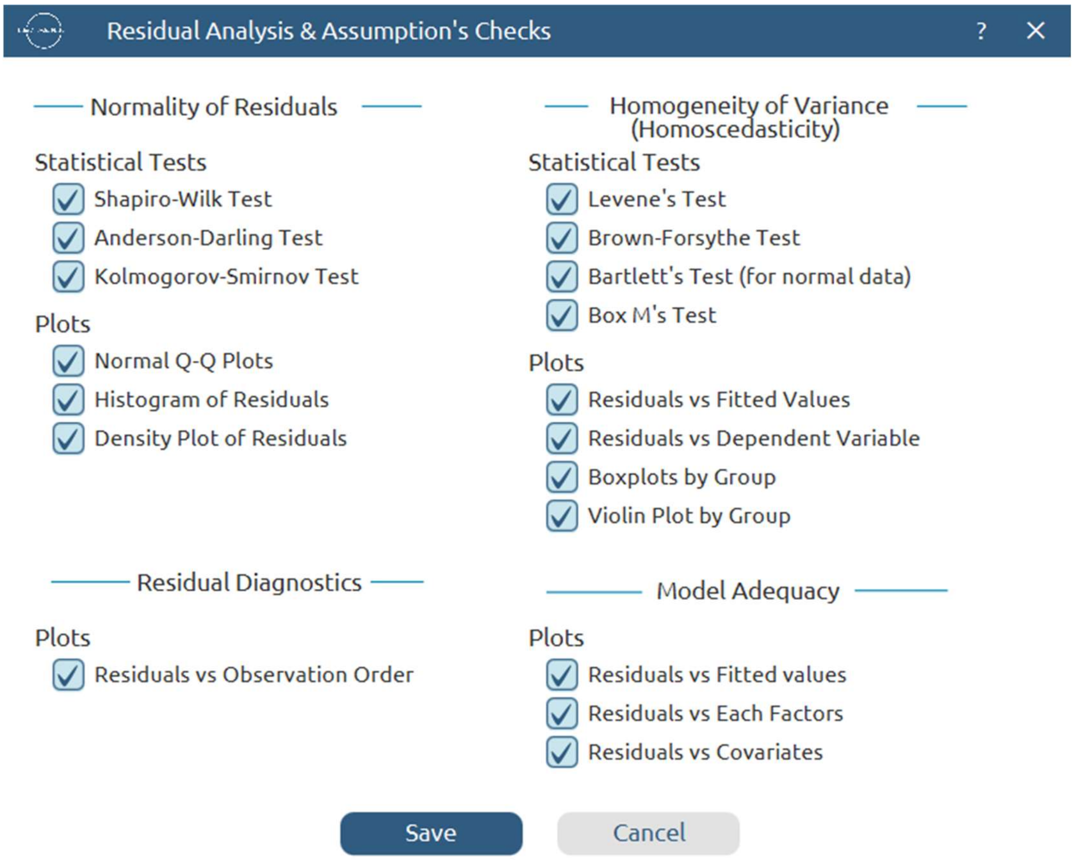
and choose

Residual Analysis & Assumption's Checks.

You get the following window:



then choose the tests you want. For example, choose the following:



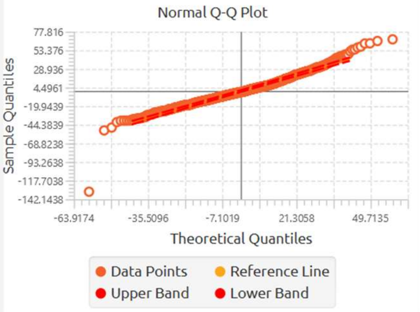
After that, click execute and you get the results (some of them below):

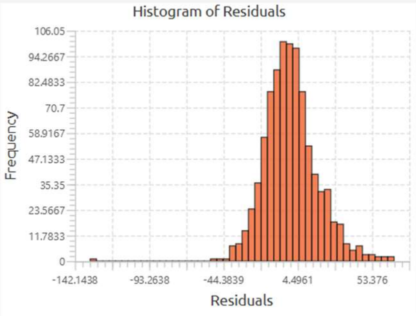
MANCOVA - RestingBP

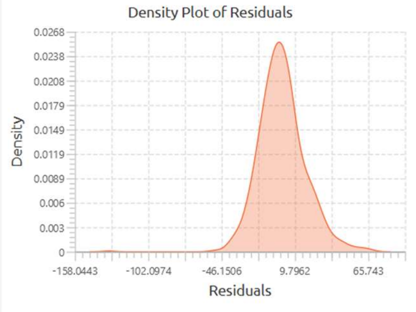
Normality of Residuals

Statistical Tests

Normality Test	Statistic	Prob	P-value	Decision
Shapiro Wilk	0.965	1.000	0.000	Reject Null Hypothesis
Anderson Darling	4.745	1.000	0.000	Reject Null Hypothesis
Kolmogorov Smirnov	0.063	1.000	0.000	Reject Null Hypothesis







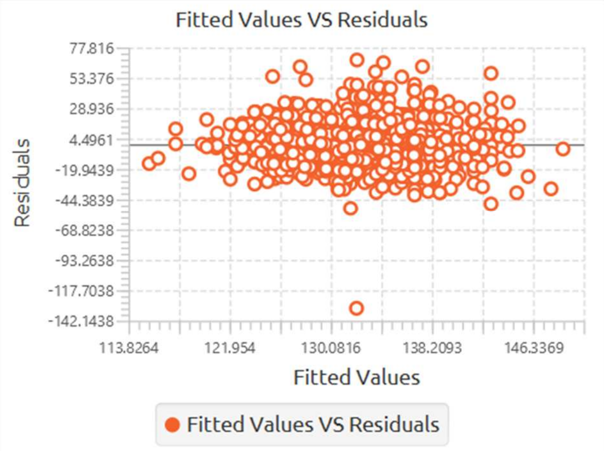
Homogeneity of Variance

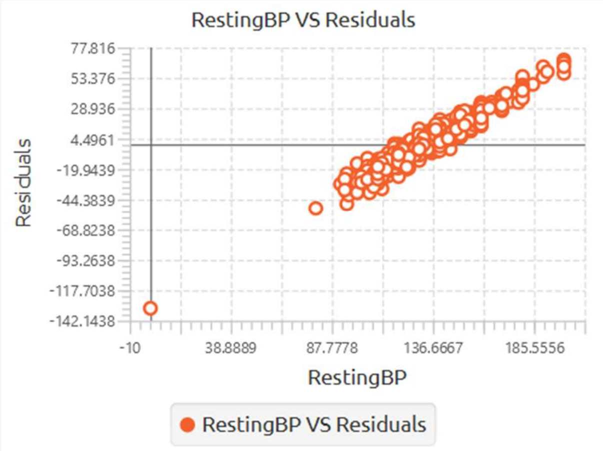
Statistical Test (Univariate)

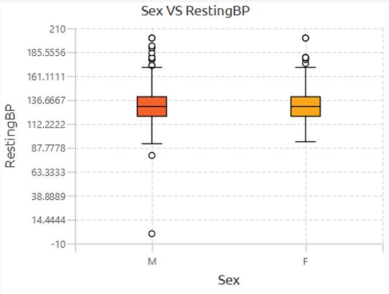
Test	F-statistic	df1	df2	P-value	Decision
Levene	1.295	7	910	0.250	Fail to Reject Null Hypothesis
Brown - Forsythe	1.158	7	910	0.325	Fail to Reject Null Hypothesis
Bartlett	17.414	7		0.015	Reject Null Hypothesis

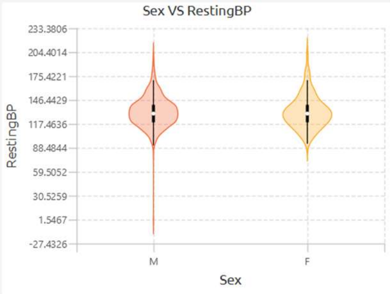
Statistical Test (Multivariate)

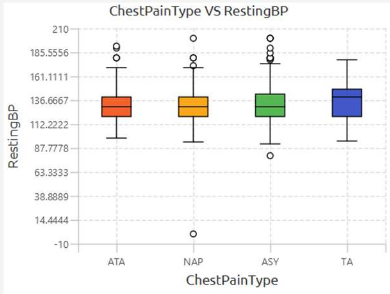
Test	Box's M	F-statistic	df1	df2	P-value	Decision
Box's M	134.574	3.094	42	18785	0.000	Reject Null Hypothesis

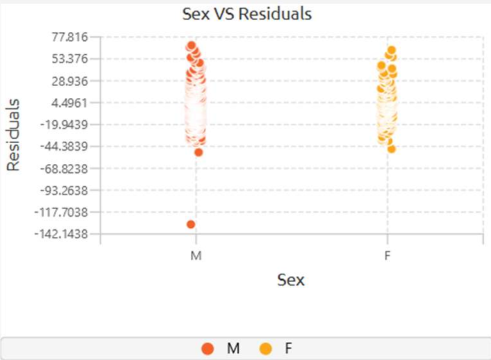


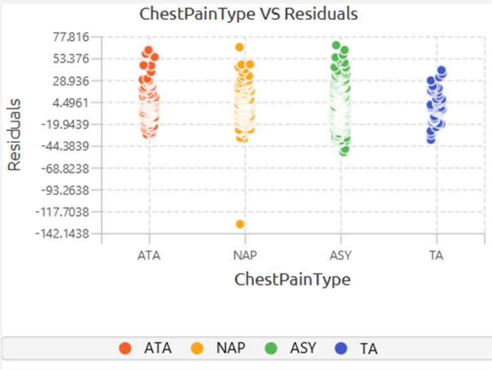


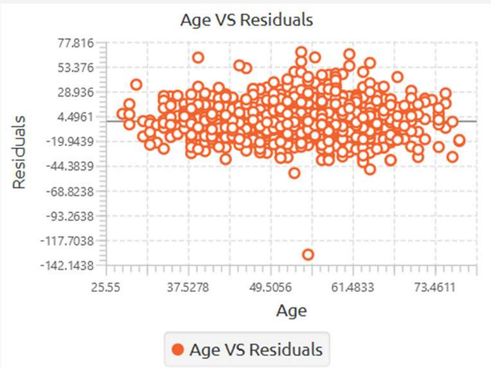










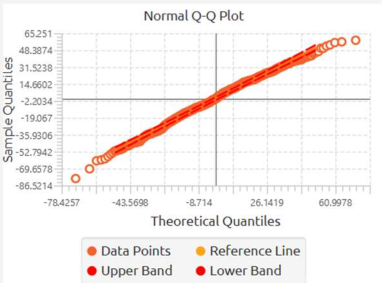


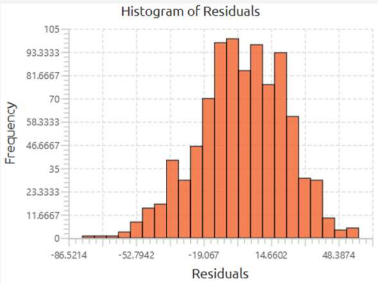
MANCOVA - MaxHR

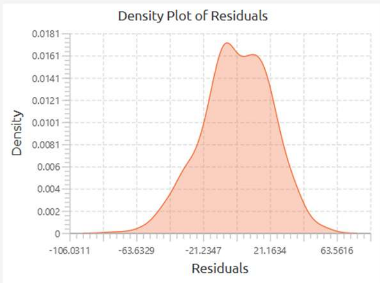
Normality of Residuals

Statistical Tests

Normality Test	Statistic	Prob	P-value	Decision
Shapiro Wilk	0.995	0.997	0.003	Reject Null Hypothesis
Anderson Darling	1.304	0.998	0.002	Reject Null Hypothesis
Kolmogorov Smirnov	0.033	0.737	0.263	Fail to Reject Null Hypothesis







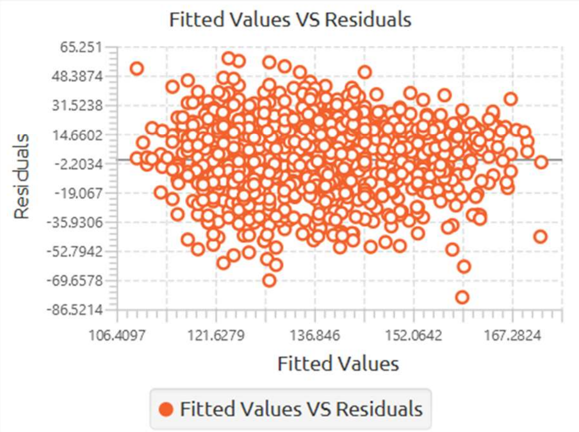
Homogeneity of Variance

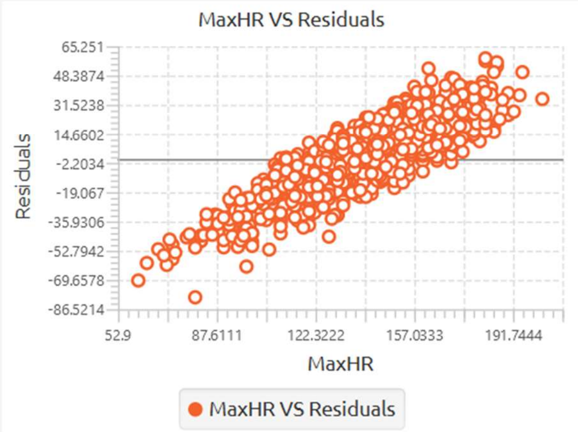
Statistical Test (Univariate)

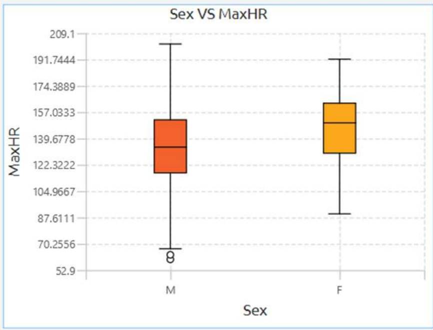
Test	F-statistic	df1	df2	P-value	Decision
Levene	1.695	7	910	0.107	Fail to Reject Null Hypothesis
Brown - Forsythe	1.661	7	910	0.115	Fail to Reject Null Hypothesis
Bartlett	10.941	7		0.141	Fail to Reject Null Hypothesis

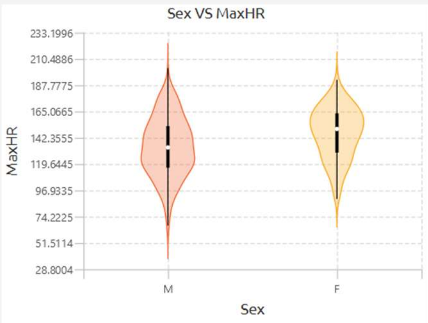
Statistical Test (Multivariate)

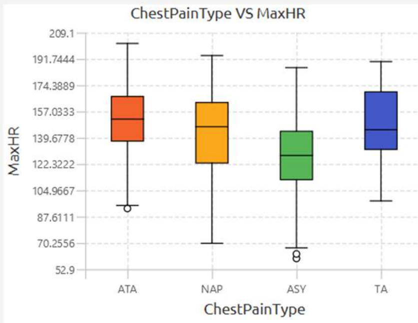
Test	Box's M	F-statistic	df1	df2	P-value	Decision
Box's M	134.574	3.094	42	18785	0.000	Reject Null Hypothesis

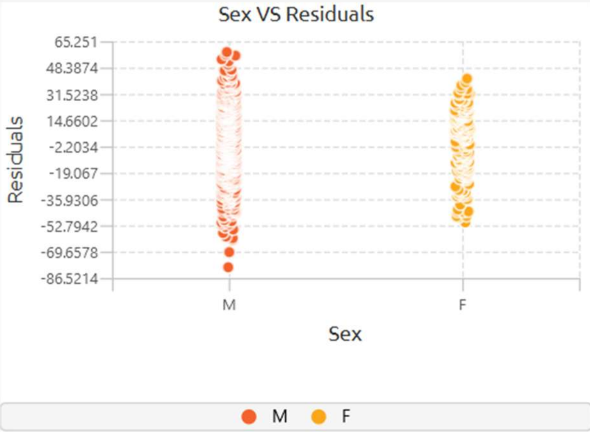


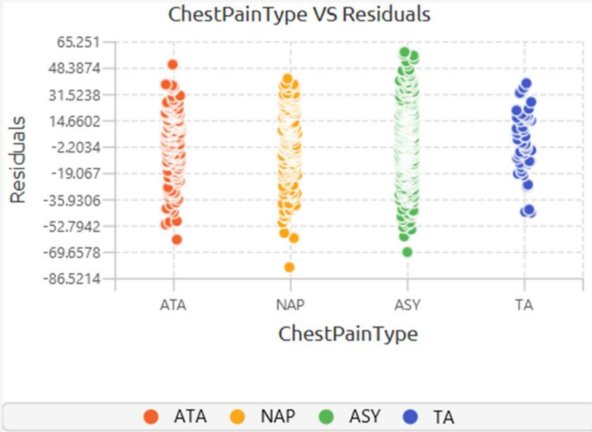


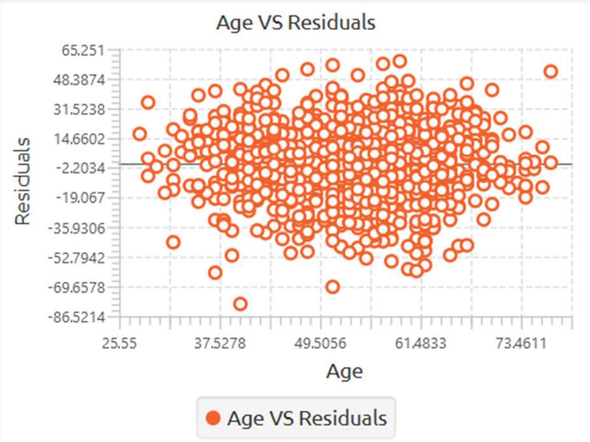










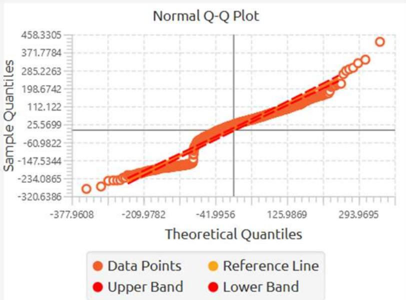


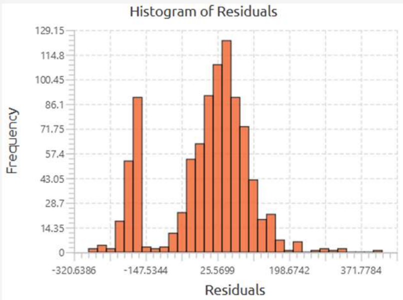
MANCOVA - Cholesterol

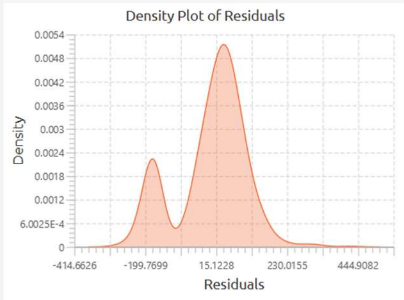
Normality of Residuals

Statistical Tests

Normality Test	Statistic	Prob	P-value	Decision
Shapiro Wilk	0.931	1.000	0.000	Reject Null Hypothesis
Anderson Darling	27.745	1.000	0.000	Reject Null Hypothesis
Kolmogorov Smirnov	0.120	1.000	0.000	Reject Null Hypothesis







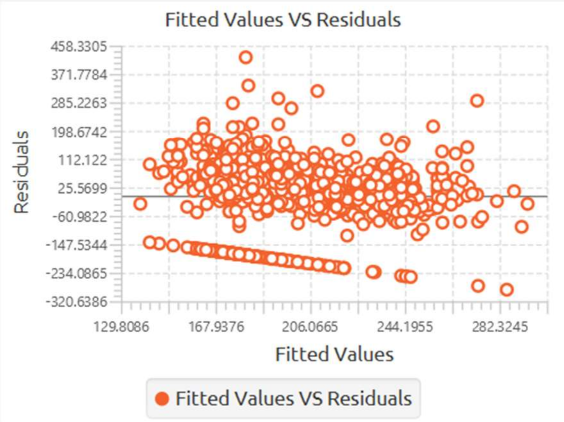
Homogeneity of Variance

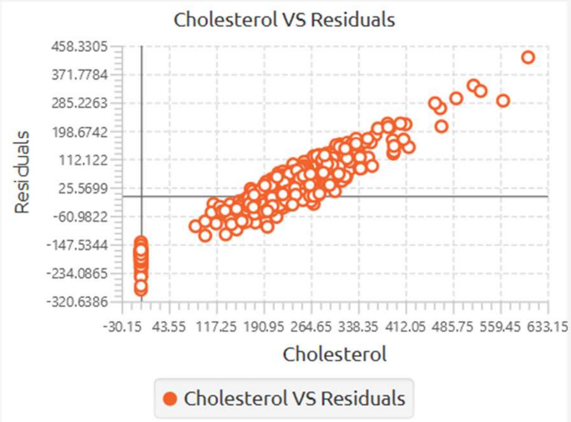
Statistical Test (Univariate)

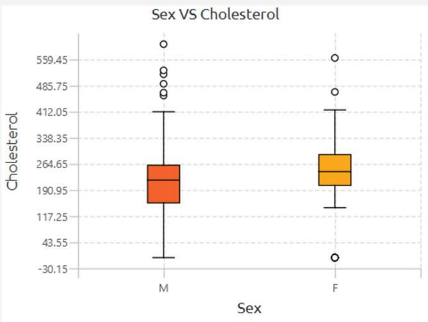
Test	F-statistic	df1	df2	P-value	Decision
Levene	16.162	7	910	0.000	Reject Null Hypothesis
Brown - Forsythe	8.829	7	910	0.000	Reject Null Hypothesis
Bartlett	83.593	7		0.000	Reject Null Hypothesis

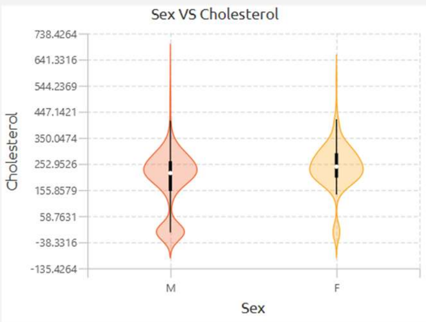
Statistical Test (Multivariate)

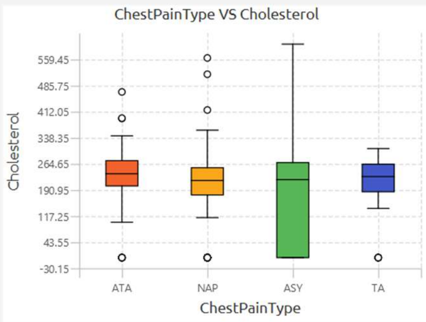
Test	Box's M	F-statistic	df1	df2	P-value	Decision
Box's M	134.574	3.094	42	18785	0.000	Reject Null Hypothesis

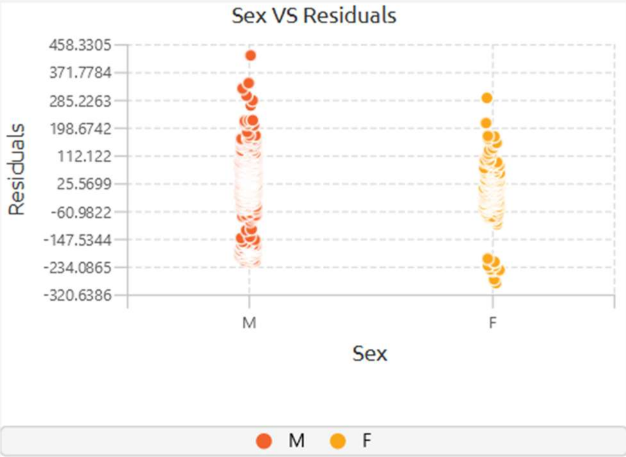


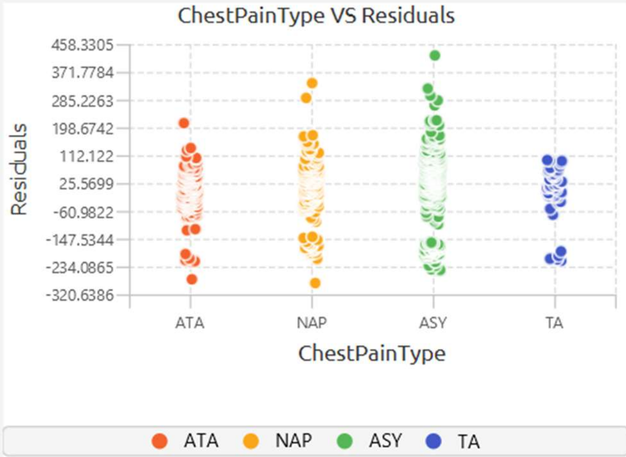


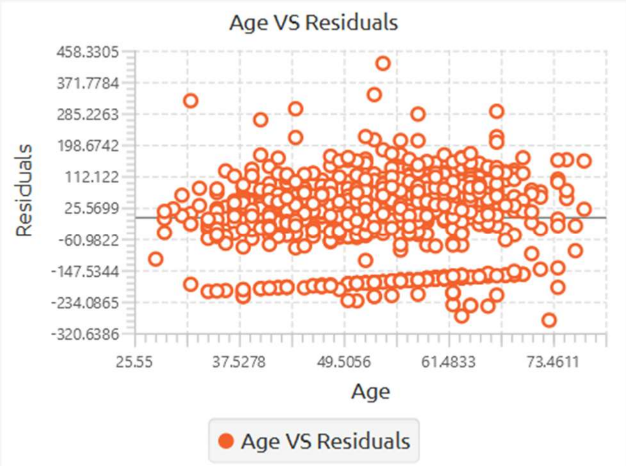







Sex VS Residuals
Residuals
458.3305
371.7784
285.2263
198.6742
112.122
25.5699
-60.9822
-147.5344
-234.0865
-320.6386
M
F
Sex
M F


ChestPainType VS Residuals
Residuals
458.3305
371.7784
285.2263
198.6742
112.122
25.5699
-60.9822
-147.5344
-234.0865
-320.6386
ATA
NAP
ASY
TA
ChestPainType
ATA NAP ASY TA


Age VS Residuals
Residuals
458.3305
371.7784
285.2263
198.6742
112.122
25.5699
-60.9822
-147.5344
-234.0865
-320.6386
25.55
37.5278
49.5056
61.4833
73.4611
Age
Age VS Residuals